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H2**Started on** Thursday, 24 September 2026, 9:19 AM**State** Finished**Completed on** Thursday, 24 September 2026, 9:19 AM**Time taken** 35 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1 | #include <stdio.h>
2 |
3 | int main() {
4 |     int n, k;
5 |     scanf("%d", &n);
6 |
7 |     int A[n];
8 |
9 |     for (int i = 0; i < n; i++) {
10 |         scanf("%d", &A[i]);
11 |     }
12 |
13 |     scanf("%d", &k);
14 |
15 |     int i = 0, j = 1;
16 |
17 |     while (i < n && j < n) {
18 |         if (i == j) {
19 |             j++;
20 |         }
21 |         else if (A[j] - A[i] == k) {
22 |             printf("1");
23 |             return 0;
24 |         }
25 |         else if (A[j] - A[i] < k) {
26 |             j++;
27 |         }
28 |         else {
29 |             i++;
30 |         }
31 |     }
32 |
33 |     printf("0");
34 |
35 |     return 0;
36 | }
37 |

```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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